

⇒ CLAMPERS (to hold) - used to calculate peak to peak value

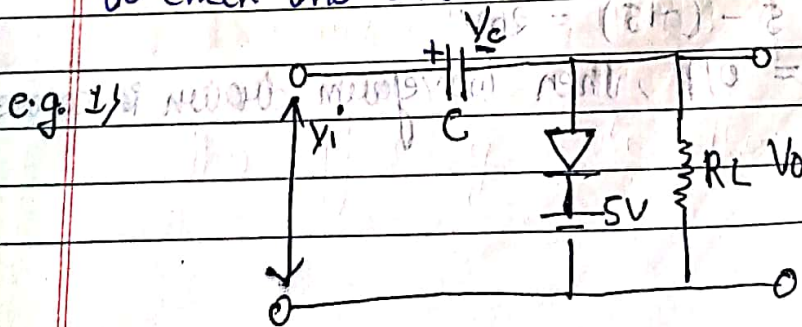
Clamper is a circuit employing a diode and a capacitor which is used to shift the signal up or down, right or left.

1. **Note** -

The output signal is never distorted; it gets only shifted.

Rules :-

- 1) Start analysing the circuit from the half cycle in which diode is forward biased.
- 2) Once the diode is forward biased, KVL is applied to find the capacitor voltage.
- 3) Once the capacitor is charged (it is always charged to the peak value of input), it will never lose its polarity or voltage.
- 4) Output waveform is plotted just below the input waveform.
- 5) Input and output voltage swings are calculated to check the answer.



▲ Step-1
Find the h.c → D → FB
+ve h.c

▲ Step-2
Find the capacitor polarity
& voltage

$$V_i - V_c = 5$$

$$V_c = V_i - 5$$

$$10 - 5 = 5V$$

$$V_i = \text{max. input}$$

$V_i(V)$

+10

-10

$V_c(V)$

-15

▲ Step-3
Vo/p during +ve h.c → D → FB
 $V_o = 5V$

▲ Step-4
Vo/p during -ve h.c → D → RB

$$V_i - V_c - V_o = 0$$

$$V_o = V_i - V_c$$

$$= -10 - 5$$

$$= -15V$$

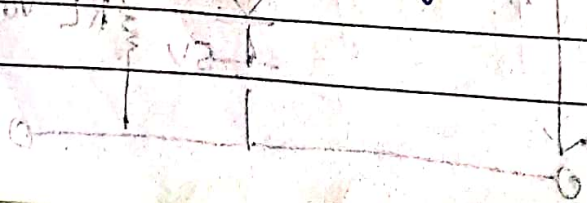
▲ Step-5

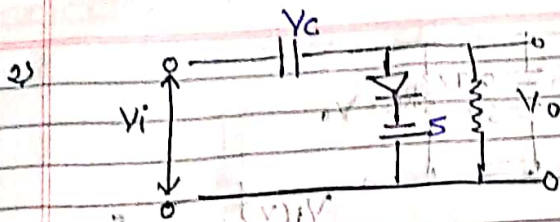
Calculate I/P & O/P voltage swing (max. - min.)

$$I/P \rightarrow 10 - (-10) = 20V$$

$$O/P \rightarrow 5 - (-15) = 20V$$

If I/P = O/P, then waveform drawn is correct





▲ Step-1
+ve h.c $\rightarrow D \rightarrow FB$

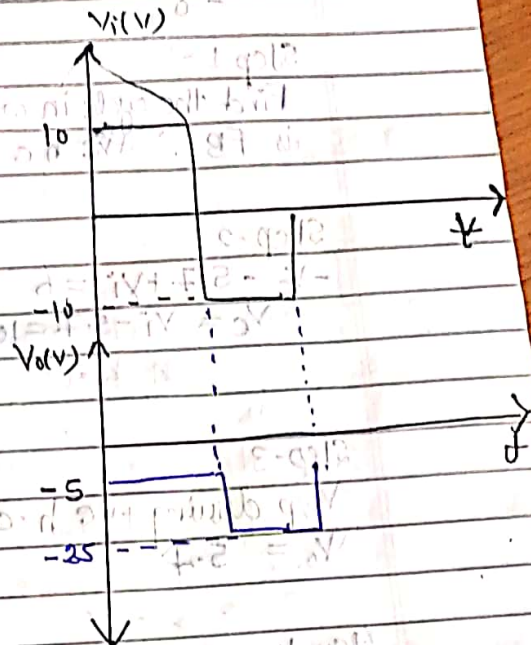
▲ Step-2
 $V_i - V_c + 5 = 0$
 $V_c = V_i + 5 = 10 + 5 = 15V$

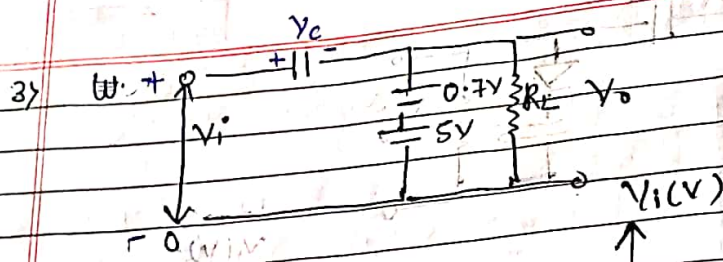
▲ Step-3
 V_o/p during +ve h.c
 $V_o = -5V$

▲ Step-4
 V_o/p during -ve h.c
 $D \rightarrow RB$
 $-V_c - V_o - V_i = 0$
 $V_o = -(V_c + V_i)$
 $-(15 + 10) = -25V$

▲ Step-5
I/P $10 - (-10) = 20$
O/P $-5 - (-25) = 20$

$\therefore$ If I/P = O/P then waveform is accurate





Step 1 -
Find the cycle in which diode is FB $\therefore$ tve h.c

Step-2
 $-V_c - 5.7 + V_i = 0$
 $V_c = V_i - 5.7 = 10 - 5.7 = 4.3V$

Step-3
 $V_{o/p}$ during tve h.c
 $V_o = 5.7$

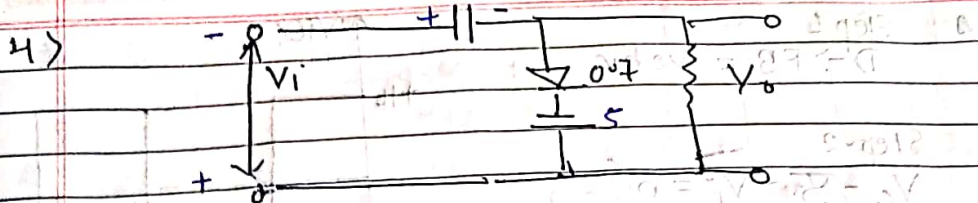
Step-4
 $V_{o/p}$ during -ve h.c
 $D \rightarrow R.B$

$$-4.3 - V_o - V_i = 0$$

$$V_o = -4.3 - 10 = -14.3V$$

Step-5

Calculate I/p & O/p voltage swing
 $I/p = 10 - (-10) = 20$
 $O/p = 5.7 + 14.3 = 20$



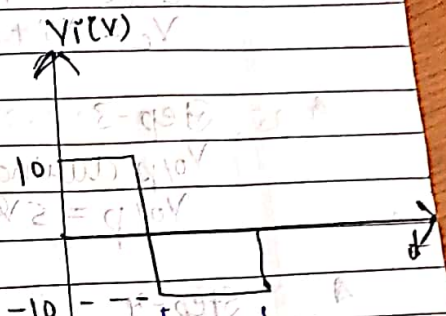
Step-1 FB + h.c

Step-2

$$-V_c - 0.7 + 5 + V_i = 0$$

$$V_c = V_i + 4.3$$

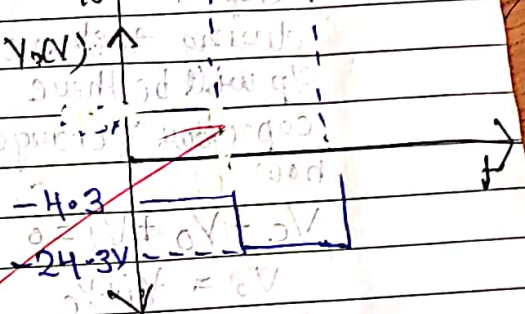
$$14.3V$$



Step-3

o/p during fve h.c

$$+0.7 - 5 = -4.3V$$



Step-4

o/p during -ve h.c

D → RB

$$\therefore -14.3 - V_o - V_i = 0$$

$$V_o = -14.3 - V_i$$

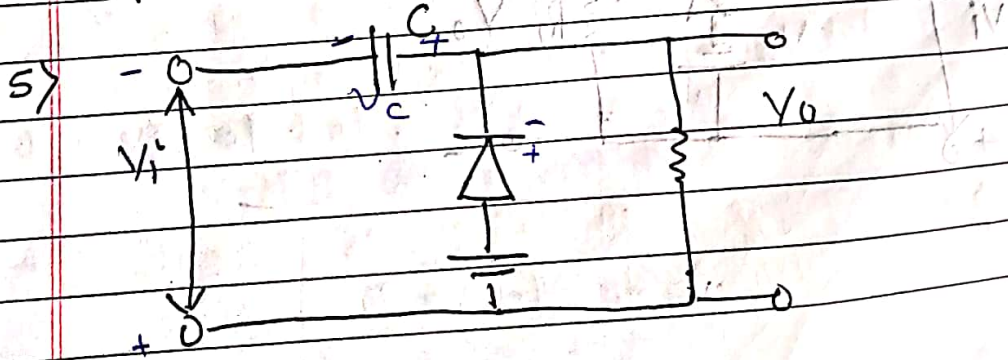
$$= -24.3V$$

Step-5 I/p & o/p voltage swing

I/p - 20

$$o/p - -4.3 + 24.3 =$$

$$20V$$



Step-1
D → FB = -ve h.c

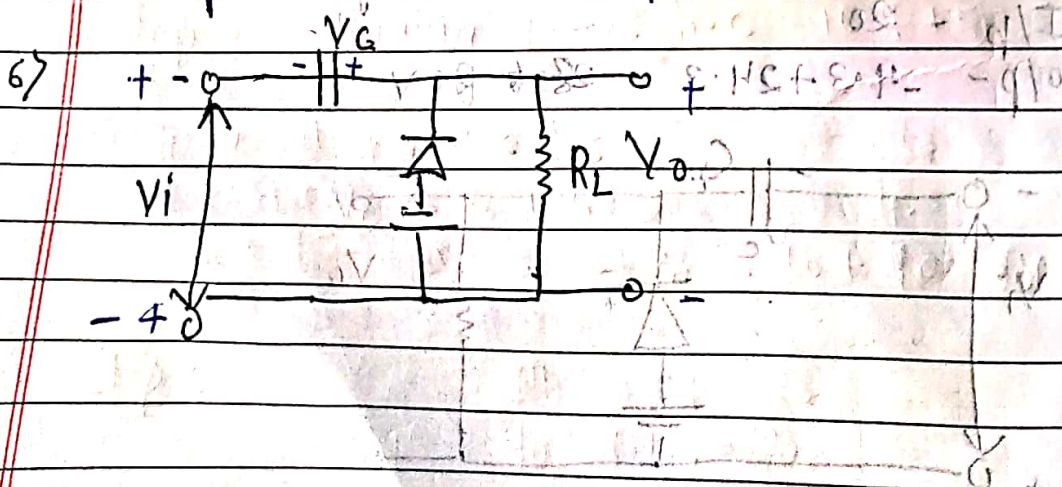
Step-2
 $V_c - V_i - V_i = 0$
 $V_c = V_i + 5 = 15V$

Step-3
Vo/p during -ve h.c
 $V_{o/p} = 5V$

Step-4
during end +ve h.c - citare is
o/p will be there bcz capacitor is charged
not changed

now
 $V_c - V_o + V_i = 0$
 $V_o = V_i + V_c$
 $10 + 15 = 25V$

Δ I/p & o/p voltage swing
I/p = $10 - (-10) = 20$
o/p = $25 - 5 = 20$ (after capacitor is charged)



Δ Step 1
D →

Δ Step 2
 $V_c + 5$
 V_c

Δ Step 3
during
 $V_{o/p}$

Δ Step
during
 $V_{o/p}$

Δ Step
I/p
I/p
0

7)

Δ

Δ

Step 1
D → FB during -ve h.c.

Step 2

$$V_c + 5 - V_i = 0$$

$$\therefore V_c = V_i - 5$$

$$10 - 5 = 5V$$

Step 3

during -ve h.c.

$$V_o = -5V$$

Step 4

during 2nd +ve h.c.

$$V_c - V_o + V_i = 0$$

$$V_o = V_c + V_i$$

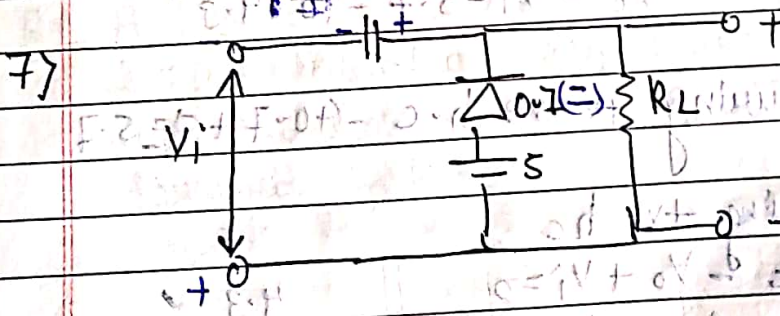
$$10 + 5 = 15V$$

Step 5

I/p & o/p voltage swing

$$I/p \quad 10 - (-10) = 20V$$

$$O/p \quad 15 - (-5) = 20V$$



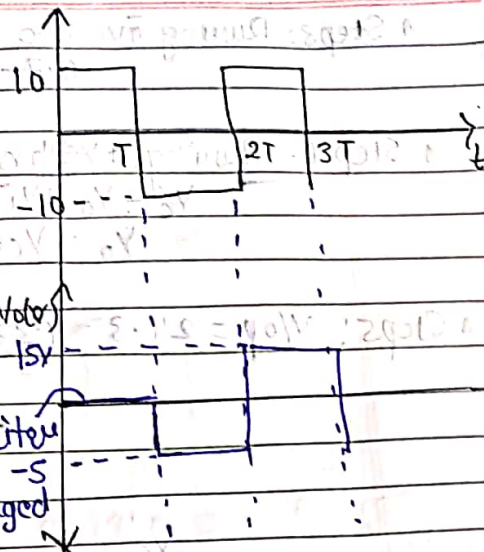
Step 1

D → FB during -ve h.c.

Step 2

$$V_c + 0.7 + 5 - V_i = 0$$

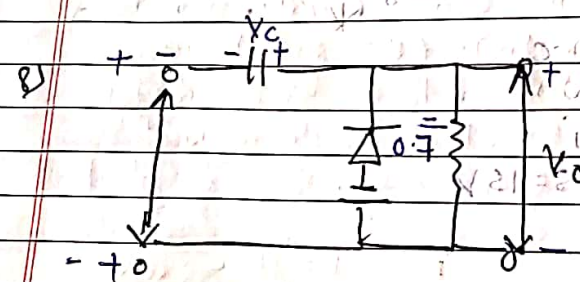
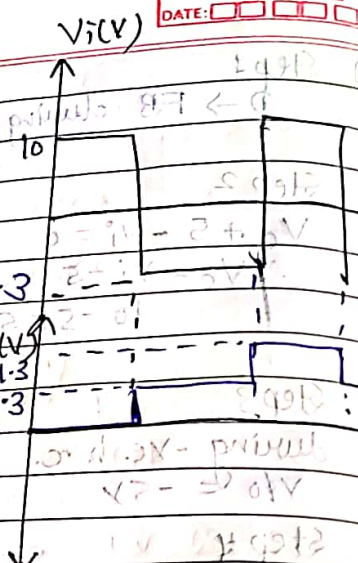
$$V_c = V_i - 5.7 = 14.3$$



Step 3: During +ve h.c
 $0.7 - 5 = 4.3V$

Step 4: During +ve h.c
 $V_c - V_o + V_i = 0$
 $V_o = V_c + V_i = 24.3$

Step 5: $V_{op} = 24.3 - 4.3 = 20$



Step 1 during -ve h.c FB

Step 2 $V_c + 0.7 + 5 - V_i = 0$
 $V_c = V_i - 5.7 = 4.3$

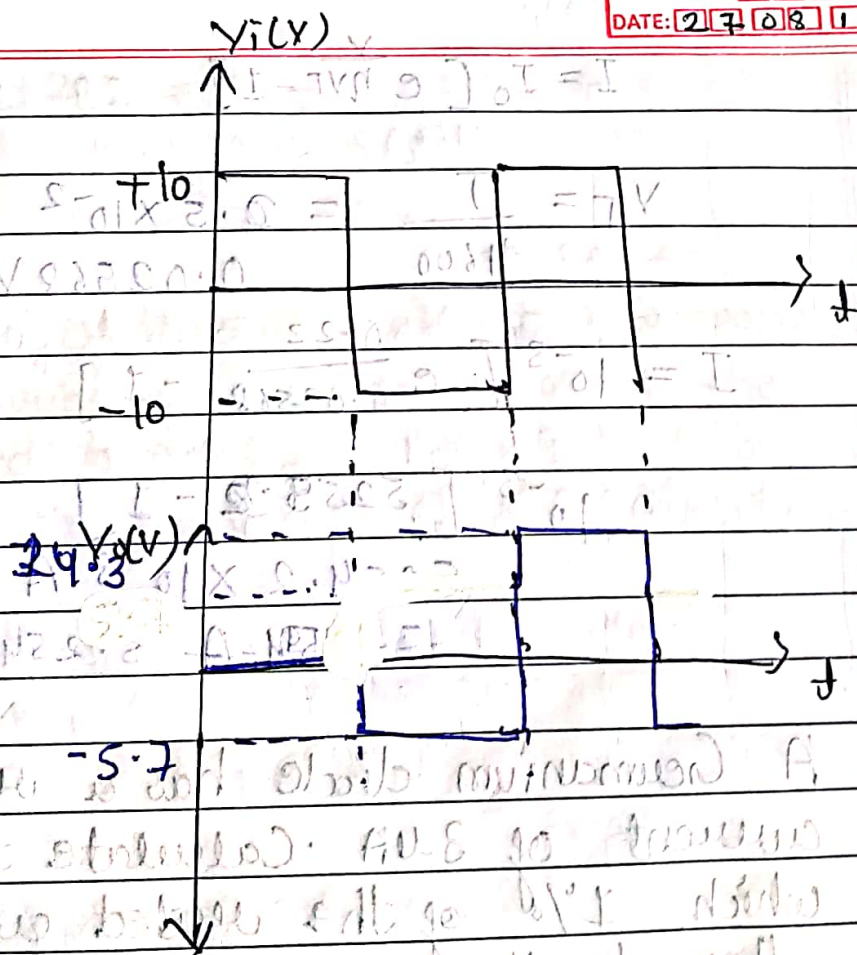
Step 3 During -ve h.c $-(0.7 + 5) = -5.7$

Step 4 during +ve h.c

$V_c - V_o + V_i = 0$
 $V_o = V_i + V_c = 10 + 4.3 = 14.3$

Step 5 $V_{op} = 24.3 + 5.7 = 30$

$V_{ip} = 10 - (-10) = 20$



Remark - If diode is connected as ∇ then waveform will shift upward and if diode is connected Δ then waveform will shift downward.